

EE 446 - TinyML - Assignment 1

Total Points: 100

Spring Quarter, 2026

Department of Electrical and Computer Engineering

University of Washington, Seattle, WA

Due: 11:59 pm (PST) on May 20th (Wednesday), 2026 via Canvas

Note:

- This assignment contains both [programming questions](#) (marked as **[Pro]**), which require writing Python code, and [discussion questions](#) (marked as **[Dis]**).
- You may discuss concepts with your classmates; however, **all Python code and answers to discussion questions must be written independently**.
- Your assignment submission must be hosted in a **public GitHub repository**. You should submit the **GitHub repository link** through Canvas as your assignment submission. Please do not submit the notebook and PDF files directly to Canvas.
- Your public GitHub repository must contain the following two files:
 - *i*) **Source code for all programming questions ([Pro])** as a **single Jupyter Notebook**, named “**#_\$_EE446_HW1_Pro.ipynb**”, where “#” and “\$” should be replaced with your first name and last name, respectively.
Clearly indicate which **[Pro]** problem each code block corresponds to, so your solution can be easily identified. Print the program output for each programming question. Include markdown cells and/or inline comments to improve readability.
 - *ii*) **Handwritten and scanned or typed answers** to all discussion questions (**[Dis]**) as a **single PDF file**, named “**#_\$_EE446_HW1_Dis.pdf**”, where “#” and “\$” should be replaced with your first name and last name, respectively.
- Your Canvas submission should contain only the public GitHub link to the repository containing the two required files above. Please make sure the link is accessible before submitting.

Problem 1: DNN and Wine Classification (80 points)

This problem is accompanied by a Python notebook named [#_EE446_HW1_Pro.ipynb](#), which contains partial code along with instructions and hints for implementation provided as comments.

Please [read through the relevant instructions carefully](#) when implementing your solution to each part.

Additionally, the corresponding [dataset](#) for this part is provided in the file named [wine.zip](#).

- (a) **[Pro] (10 pts)** Implement a baseline DNN model for wine classification using the UCI wine dataset. Report training accuracy, test accuracy, and float32 model size.
- (b) **[Pro] (20 pts)** Quantize the baseline model using:
 - Dynamic Range Quantization
 - Full Integer Quantization
 - Float16 Quantization

Report model size and classification performance (accuracy, classification report, confusion matrix) for each quantized model.

- (c) **[Pro] (20 pts)** Apply pruning to the baseline model and evaluate the size and performance of the pruned model.
- (d) **[Pro] (20 pts)** Apply output-based Knowledge Distillation using the baseline model as the teacher and a smaller student model. Evaluate and report performance and model size of the student.
- (e) **[Pro/Dis] (10 pts)** Can you further reduce model size beyond parts (b), (c), and (d)? If yes, implement your proposed method and compare results. If not, explain why.

Problem 2: Exploring Edge Impulse (20 points)

1. **[Dis] (5 pts)** What types of “Learning Blocks” are available under the free-tier Edge Impulse account (e.g., supervised learning—classification and regression, unsupervised learning—clustering, self-supervised learning, etc.)?
For each type, provide at least one example model that can be implemented using the free tier.
2. **[Dis] (5 pts)** List all the types of layers that can be used under the “Classifier” section in the free-tier Edge Impulse. Briefly explain the purpose of each layer identified.
3. **[Dis] (5 pts)** What types of model compression options are available under the “Deployment” section in the free-tier Edge Impulse? List all available methods.
4. **[Dis] (5 pts)** What input modalities are supported for synthetic data generation in the free-tier version of Edge Impulse? For each modality, list the synthetic data generation methods/algorithms provided. Briefly explain why synthetic data is important in machine learning applications.